

implies that an element with a context-specific tag cannot appear as the outermost element of a TLV encoding.

A.2.3. Anonymous Tags

A special "anonymous tag" is used to denote TLV elements that lack a tag value. Such a TLV element is referred to as an anonymous element.

A.2.4. Canonical Ordering of Tags

Where a distinguished ordering of tags is required (e.g. for the purposes of generating a hash or cryptographic signature of elements within a structure), the following ordering rules SHALL be used:

- Anonymous tags SHALL be ordered before all other tags.
- Context-specific tags SHALL be ordered before profile-specific tags.
- Context-specific tags with numerically lower tag values SHALL be ordered before those with higher tag values.
- Profile-specific tags with numerically lower Vendor IDs SHALL be ordered before those with higher Vendor IDs.
- Profile-specific tags with the same Vendor ID, but numerically lower profile numbers SHALL be ordered before those with higher profile numbers.
- Profile-specific tags with the same Vendor ID and the same profile numbers but numerically lower tag numbers SHALL be ordered before those with higher tag numbers.

The ordering rules SHALL apply to elements at the same level within a container.

A.3. Lengths

Depending on its type, a TLV element may contain a length field that gives the length, in octets, of the element's value field. A length field is only present for string types (character and octet strings). Other element types either have a predetermined length or are encoded with a marker that identifies their end.

A.4. Primitive Types

The Matter TLV format supports the following primitive types:

- Signed integers
- Unsigned integers
- UTF-8 Strings
- Octet Strings
- Single or double-precision floating point numbers (following [IEEE 754-2019](#))
- Booleans